



Fig. 3: Actual-size PCB pattern of the circuit for the infrared burglar alarm

PARTS LIST

Semiconductors:

IC1	- CD4060 ripple-carry binary counter
IC2	- UM3561 tone generator
T1, T2	- BC547 npn transistor
T3	- SL100 medium-power npn transistor
D1	- 1N4148 diode
LED1	- 5mm LED
LED2	- White LED, 0.5W
ZD1	- 5.1V zener diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

R1	- 100-kilo-ohm
R2, R10	- 1-kilo-ohm
R3	- 10-mega-ohm
R4	- 1-mega-ohm
R5-R7	- 470-ohm
R8	- 220-kilo-ohm
R9	- 100-ohm, 0.5W

Capacitor:

C1	- 0.22 μ F ceramic disk
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Miscellaneous:

LDR1	- Light-dependent resistor
CON1	- 3-pin connector
BATT.1	- 9V battery
LS1	- 8-ohm, 0.5W speaker
	- PIR motion-sensor module
	- 2-pin connector terminal for battery

pin 12 of IC1. During day time, LDR1 has low resistance in comparison with R1, thereby making reset pin high, which, in turn, prevents IC1 from oscillating during day time.

At night, the situation gets reversed. LDR1 has very high resistance in comparison with R1, so reset pin 12 of IC1 gets pushed to ground potential.

Light-emitting diode LED1 indicates activation of timer IC1. Capacitor C1 and resistor R3 form oscillating components of timer IC1. With the given values of C1 and R3, output at pin 1 of IC1 becomes high after approximately three hours from the time the reset pin went low.

High output does two things. It drives transistor T1 to saturation and sets pin 11 of IC1 high through forward-biased diode D1. IC1 stops oscillating. Its output Q12 remains high till IC1 gets reset in the morning due to low resistance of LDR1.

When T1 is in the state of saturation, the emitter is at 9V DC. The regulator circuit formed by R6 and zener diode (ZD1) provide regulated 5.1V DC at cathode of zener diode ZD1, which is connected to the PIR sensor.

The sensor is now active and ready to detect human presence up to a distance of a few metres (depending on the range setting of the preset) and up to an angle of 45°. When it detects human presence, its output turns high for the period set to activate the alarm and to switch on the white LED (LED2).

The alarm is built around tone generator IC UM3561 (IC2). Resistor R8 (220k) determines the frequency

of the oscillation of IC2. Its output is amplified by transistor T3. The alarm sound can be heard through the speaker. UM3561 can generate different alarm tones; here it is designed to sound a police siren.

When the PIR sensor gives high output, transistor T2 drives the 0.5W white LED. It gives sufficient light to catch the intruder.

Construction and testing

An actual-size, single-side PCB of the infrared burglar alarm is shown in Fig. 3 and its component layout in Fig. 4.

After assembling the circuit, enclose it in a suitable box. The unit should be fixed on an outside wall with the PIR sensor facing downwards at an angle of 45°. The sensor should be about two metres above ground level to get maximum coverage. LDR1 should get ambient light for proper operation of the timer. Light from LED2 should not fall on LDR1, as this could deactivate the timer. ●



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